[INCH-POUND]
A-A-59218
May 18, 1998
SUPERSEDING
OO-C-2763
May 30, 1991

COMMERCIAL ITEM DESCRIPTION

CRANES, HYDRAULIC, CARRIER MOUNTED

The General Services Administration has authorized the use of this commercial item description by all Federal agencies.

1. SCOPE. This commercial item description (CID) covers diesel-engine-driven, truck mounted, full revolving, hydraulically operated cranes.

2. SALIENT CHARACTERISTICS.

2.1 Description. The crane shall be a commercial model, revolving superstructure with cab and a powered hydraulic telescoping boom, mounted on a pneumatic tired, diesel-engine-driven carrier. The crane shall comply with the requirements of the current mobile hydraulic crane standards.

2.2 Standard commercial product. The hydraulic cranes shall, as a minimum, be in accordance with the requirements of this commercial item description and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this commercial item description but which are a part of the manufacturer's standard commercial product, shall be included in the hydraulic cranes being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3810

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2.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

2.4 Safety. The crane shall comply with Occupational Safety and Health Administration (OSHA) regulations and Power Crane and Shovel Association (PCSA) standard. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to the operating personnel, shall be guarded or insulated. The principal platform walking surfaces shall be of anti-skid type. Ladders, steps, and handholds shall be provided on the sides of the crane cab or superstructure. The entrance and exit from the cab shall be unhampered and nonhazardous. Engine cooling fans shall have heavy grille or ring type guards. The crane counterweight, boom point, and hook block cheek plates shall be painted with stripes. Stripes shall be 2 inches (51 millimetre (mm)) wide, with 2-inch (51 mm) spacing extending at a 45-degree angle. Stripes shall be black.

2.5 Performance. The crane shall be capable of independently and simultaneously lifting and rotating at a speed of not less than 2 revolutions per minute. The crane, without load, shall be capable of attaining and maintaining a travel speed, on level paved roads, of not less than 40 miles per hour (64.4 kilometre per hour). The crane, without load, shall be capable of traveling a 20 percent grade on paved roads or hard-packed surfaces without engine stall; and shall be capable of negotiating 15 percent side slopes.

2.5.1 Lifting capacities. The crane shall have lifting capacities of not less than those specified at the radii and boom lengths specified (see 6.2).

2.5.2 Stability. With lubricating and cooling systems full, fuel tank(s) not less than half full, with crane resting on firm, level, and uniform supporting surface, with outriggers unsupported, and with the boom fully retracted, the crane shall conform to the following requirements:

a. With the longitudinal axis of the rotating superstructure of the crane at 90 degrees to the longitudinal axis of the carrier, and the boom retracted and set at the minimum recommended radius, the total load on all wheels on the side of the carrier under the boom shall be not less than 15 percent of the total weight of the crane.

b. With the longitudinal axis of the rotating superstructure of the crane in line with the longitudinal axis of the carrier, in either direction, and the boom retracted and set at the minimum recommended radius, the total load on all wheels under the lighter loaded end of the carrier shall be not less than 15 percent of the total weight of the crane.

2.5.3 Power load lowering. Power load lowering by hydraulic control shall be provided for the main and auxiliary hoists to facilitate controlled load lowering or lifting at any boom lengths or elevation. The power load lowering means shall provide positive control of a single line load equal to the maximum rated crane capacity divided by the recommended number of parts of hoist line reeving.

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2.5.4 Hoist and rotation. The available line pull and speed shall be as rated in accordance with Power Crane and Shovel Association (PCSA) standard for the operational requirements of the crane hoisting drums. The power transmission system used to rotate the superstructure shall be a hydraulic mechanical arrangement with a continuous swing rotation angle of not less than 360 degrees. Swing speed within this range shall be smooth and constant, and adequate control shall be provided in the hydraulic circuit to ensure smooth starting and stopping. The crane shall simultaneously lift and rotate throughout the rated working area. The main hoist shall be of the two-speed type. The auxiliary winch shall be manufacturer's standard. An auxiliary boom point sheave shall be furnished.

2.5.5 Rated load capacities chart. A permanent chart indicating all the rated load capacities on each boom reach and radii over the side and over the rear, with outriggers, shall be posted in the crane cab near the operator. The chart shall also be posted on the outside of the crane to be legible from ground level. The rated load capacity chart shall reflect the manufacturer's standard commercial rating for the size and model crane furnished, and not be limited to the minimum required capacities specified.

2.6 Crane. The crane assembly, designed for the carrier mounting, shall consist of a fully revolving superstructure, hydraulically operated telescoping type boom, main and auxiliary winches with wire rope, hook block, cab, controls, and necessary accessories.

2.6.1 Superstructure. The superstructure shall provide stable retention of the boom in all positions.

2.6.2 Boom. The boom shall be of the hydraulically operated telescoping type. The boom shall be capable of elevation from the horizontal to a vertical angle of not less than 72 degrees. A digital or dial type boom angle indicator shall be furnished. An anti-two-block device with audio-visual warning and function lockout shall be installed to prevent the hook block from contacting the boom tip. An auxiliary sheave shall be provided to permit the use of auxiliary hoist attached to main boom (rooster sheave).

2.6.2.1 Boom drift. Boom drift, the vertical dropping of the boom due to inadvertent retraction of the boom hoist or boom extension cylinders, shall be kept to a minimum. Drift shall take place in gradual, imperceptible steps or degrees. The crane load, under all operating conditions, shall not drop in a sharp or sudden manner at any time during crane operation.

2.6.3 Boom jib. A self-storing boom jib or swing-away boom extension shall be provided. Length of the boom jib or extension shall be as specified (see 6.2). Boom jib shall be provided with the necessary risers, overhaul ball with swivel hook, and ropes. Boom jibs shall be readily attachable to the tip of the boom with the capability of withstanding the maximum jib load without malfunctions.

2.6.4 Hook block. The rated capacity of the crane hook block shall be not less than the rated capacity of the crane for which it is furnished. The hook shall be of the swivel type with an ultimate breaking strength not less than four times the maximum rated capacity of the crane and

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shall include a safety latch. The block assembly shall have adequate weight to ensure payout of the hook in any position. The hook block shall permit disassembly for visual and nondestructive testing. All hook block(s) shall be equipped with safety latches. The hook shall not be painted.

2.6.5 Rope. Load hoist ropes shall be improved plow steel, Independent Wire Rope Center (IWRC), with a safety factor of not less than 3.5. The main and auxiliary winches shall be furnished with the manufacturer's standard length rope. The contractor shall provide a signed certification with each crane identifying the crane serial number, wire rope manufacturer, type of wire rope construction, type of wire rope material, and the minimum breaking strength of the wire rope provided. Load hoist ropes used on auxiliary winches shall be nonrotating type.

2.6.6 Sheaves. Sheaves shall have a pitch diameter not less than 18 times the rope diameter. The hook block pitch diameter shall be not less than 16 times the rope diameter. Sheaves shall be adequately guarded to prevent the wire rope from leaving the sheave during operation. All sheaves shall be smooth and the nominal groove diameter shall be not greater than the nominal diameter of the wire rope furnished by more than 0.10-inch (2.54 mm). Steel sheaves shall be furnished. All sheaves shall be equipped with grease fittings.

2.6.7 Hydraulic system. Hydraulic line circuit and filtration shall be in accordance with SAE J931. Means shall be provided in the hydraulic system for bleeding all air trapped in the system. All pressure hydraulic hoses and fittings shall be capable of withstanding a burst pressure of not less than four times the maximum operating pressure, and a proof pressure of not less than two times the maximum operating pressure.

2.6.8 Load moment device. A load indicating and warning device of the load moment type shall perform the following functions while on outriggers.

a. Provide a dial or digital reading of hook load in pounds (kilogram (kg)).
b. Provide a dial or digital reading of radius in feet (metre (m)).
c. Provide a dial reading of load moment as a percentage of rated capacity.
d. Provide a visual warning (amber light) when load moment reaches 90 percent of capacity and visual (red light) an audible warning at 100 percent of capacity.
e. Provide function kickout (load may be lowered but not raised) when load moment reaches 100 percent of rated capacity.

The device shall be capable of manually selecting parts of line and boom length. The device shall conform to SAE J159, J375, and J376.

2.7 Cabs. The cabs shall be fabricated from sheet steel or fiberglass. The cab shall have sufficient windows and glazed doors to permit not less than 270-degree vision for the driver or operator in normal position. All glass shall be safety type. Windows shall be arranged for easy opening and removal when required. All cab doors, whether of the sliding or swinging type, shall be adequately restrained from accidentally opening or closing when the crane is traveling or operating. All doors shall be provided with means for securely locking in the closed position. Adjustable and comfortable seats shall be provided in each cab and located for maximum

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unobstructed visibility of the operator. Both cabs shall be equipped with heater and defroster, air or electric operated windshield wiper, electric or air horn. A positive swing lock for traveling shall be provided. The carrier cab shall be provided with West Coast-type outside rearview mirrors on each side.

2.8 Controls. All hoist, crane controls, lockouts, and indicators shall be located within easy reach of the operator in the crane cab. All controls shall be clearly marked as to the use and function.

2.9 Carrier. Unless otherwise specified (see 6.2), the crane carrier shall be the manufacturer's standard width for the capacity crane furnished. The number of axles shall be the manufacturer's standard for the capacity crane furnished with all rear axles to be driven. Power-assisted steering for the front wheels shall be provided. If crane furnished has more than two rear axles, not less than two rear and one front axle shall be driven.

2.9.1 Transmission. The crane carrier shall be equipped with the manufacturer's standard transmission for the size crane specified. The transmission shall be equipped with a neutral safety start switch.

2.9.2 Brakes. The crane carrier shall be provided with the manufacturer's standard, foot controlled, service brakes applied to all wheels. The service brakes shall be capable of bringing the crane to a complete safe stop within the distance specified in SAE J1152. A parking brake capable of holding the crane, on a longitudinal slope of not less than 20 percent without slippage, shall be provided.

2.9.3 Tires. Tires shall be of the tube or tubeless type with traction type tread. Tire ratings shall conform to TRA recommendations for the type and size of tires furnished. When specified (see 6.2), tires shall be high flotation type.

2.9.4 Lights and reflectors. The following lights and reflectors shall be furnished and installed and shall be powered by a 12-volt electrical system:

a. The sealed beam headlights (mounted on carrier).
b. Taillights.
c. Stoplights.
d. Clearance lights.
e. Turn signal lights.
f. Instrument panel light (both cabs).
g. A weatherproof automotive type of spotlight or floodlight.
h. Emergency 4-way flashing lights.
i. Interior dome lights (both cabs).

All carrier lights, reflectors, and wiring shall conform to Department of Transportation Motor Carrier Safety Regulations.

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2.9.5 Engine(s). The crane engine(s) furnished, single or dual, shall be of the diesel type. The diesel engine(s) shall start within 5 minutes and be ready for full load operation within 15 minutes in any ambient temperature from + 120 degrees Fahrenheit (°F) to - 20 °F (102 degrees Celsius (°C) to - 38 °C)). When a fluid priming system is required, it shall be of the measured shot type with storage capacity at not less than 12 fluid ounces (355 millilitre). The engine(s) shall be furnished complete with not less than the following accessories:

a. An instrument panel complete with a lubricating oil pressure gage, a fuel oil gage, a cooling liquid-temperature indicator, a voltmeter or ammeter, a tachometer, an hour meter (may be located in engine compartment), and a speedometer with recording odometer.

b. Fuel tank(s) with sufficient capacity for 8 hours normal operation.

c. 12-volt or 24-volt electric cranking system.

d. When specified (see 6.2), battery(s) shall be dry charged without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture.

e. Charging alternator(s) with a capacity rating compatible to the system.

f. Cooling liquid high temperature safety device of the warning type.

g. Lubricating oil safety device.

h. Fuel oil level gage.

i. An engine housing so designed as to facilitate the maintenance of the engine.

j. Transmission temperature gage (automatic transmission only).

2.9.6 Outriggers. Not less than four hydraulically powered outriggers, capable of being extended not less than 3 inches (76 mm) below ground level, shall be furnished. The outriggers shall have the ability to raise the truck cranes off the ground from travel position, when necessary to compensate for ground surfaces having irregular contours. Outriggers in the stowed position, ready for travel, shall be not greater than the maximum allowable overall width. Devices to accurately determine when the cranes are level shall be installed in a position convenient to the outrigger controls. A front mounted outrigger shall be furnished on all cranes 30 tons (27 216 kg) and larger. Outrigger pads shall be removable for extended travel. Outrigger pads shall be stowed on carrier.

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2.10 Attachments. The following attachments in type and size shall be furnished as specified (see 6.2):

a. Clamshell bucket and tagline.
b. Towing hook.
c. Towing winch.

2.11 Radio noise suppressors. When specified (see 6.2), the crane shall be equipped with radio noise suppressors. Electromagnetic radiation from the crane shall be within the limits of SAE J551.

2.12 Special tools and toolbox. A closed compartment or lockable metallic toolbox shall be provided to carry tools and accessories. Those specific tools that are unique to the crane furnished and are required for operation shall be furnished. The toolbox shall be securely fastened to the unit in a protected and accessible location.

2.13 Lifting and tiedown attachments. When specified (see 6.2), the cranes shall be equipped with lifting and tiedown attachments. A nonferrous transportation plate shall be provided and mechanically attached to the cranes. Transportation shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the cranes on the carrier when shipped.

2.14 Identification plate. An identification plate will be furnished by the contracting officer for each crane. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each crane in a conspicuous place with nonferrous screws, rivets, or bolts. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

2.15 Instruction plates. The cranes shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts.

2.16 Stenciling. The gross weight of each crane, with equipment, shall be stenciled on each side of the crane in such a manner as to be readily discernible to dock and military personnel. The prescribed tire pressure shall be stenciled on the carrier fenders by block or stencil-type letters not more than 1-inch (25 mm) high.

2.17 Lubrication. Means for lubrication shall be in accordance with the manufacturer’s standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment,

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1,000 pound-force per square inch (psi) (6 895 kilopascal) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

2.18 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning and before any corrosion product or other contamination can results, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils (0.0635 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

2.19 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10, and all other requirements of this commercial item description including form, fit, and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for changes to the document.

3. REGULATORY REQUIREMENTS.

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Source of documents.

6.1.1 The Federal Acquisition Regulation (FAR), Department of Transportation Motor Carrier Safety Regulations, and OSHA Standards are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.1.2 ASTM Standards are available from the American Society of Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.1.3 PCSA Standard is available from the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 East Wisconsin Avenue, Milwaukee, WI 53202.

6.1.4 SAE Standards are available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

6.1.5 TRA Yearbook is available from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Crane lifting capacities, with and without outriggers, at specific radii, and boom lengths required (see 2.5.1).
c. Length of boom jib or extension required (see 2.6.3).
d. When carrier width is other than manufacturer's standard (see 2.9).
e. When high flotation type tires are required (see 2.9.3).
f. When batteries are other than as specified (see 2.9.5d).
g. Attachments required (see 2.10).
h. When radio noise suppressors are required (see 2.11).
i. When lifting and tiedown attachments are required (see 2.13).
j. Color of finish coat (see 2.18).

6.3 Supersession data. This CID replaces Federal Specification OO-C-2763 dated May 30, 1991.

6.4 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	NOMENCLATURE
3810-00-255-6345	Crane, Truck Mounted
3810-01-038-8315	Crane, Truck Mounted

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MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activities:
Navy - MC
Air Force - 84
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Navy - YD1

(Project 3810-0008)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59218	**2. DOCUMENT DATE (YYMMDD)** 980518

3. DOCUMENT TITLE

CRANES, HYDRAULIC, CARRIER MOUNTED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. E. PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC CODE 15E2K 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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